

# Reply to RFQ / BOM quotation — T-2SJ3W1113248A

**Product:** Racecar-35 Rev D integrated controller

**Quantity:** 5 units · turnkey PCB assembly (parts + assembly + bare PCB)

**Date:** 29 September 2026 · **Quote reference:** T-2SJ3W1113248A

Please apply the decisions below and issue the **revised quotation**. The

corrected BOM to match from is attached:

Racecar-RevD-BOM-pcbway-assembly-REV1.csv (six part numbers corrected, plus

fitting instructions for U1 and U2 in the Comment column).

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## 1. Answers to the three "substitute OK?" questions

| Item | Original                 | Your proposed substitute | Our answer                                                                                                                                                                                                                     |
|------|--------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 21   | GRM21BR71H103KA01L (C10) | CL21B103KBANNNC          | <b>APPROVED</b> — same value: 10 nF, 50 V, X7R, 0805                                                                                                                                                                           |
| 22   | GRM21BR71H225KA01L (C11) | CL21A225KBQNNNE          | <b>NOT APPROVED</b> — this is X5R (85 °C). The original is X7R (125 °C) and this part sits on the 3.3 V MCU rail. Please use <b>CC0805KKX7R9BB225</b> (Yageo, LCSC C125847) or <b>CL21B225KBYNNNE</b> (Samsung, LCSC C2762602) |
| 35   | GRM21BR71H102KA01L (C18) | CC0805KRX7R9BB102        | <b>APPROVED</b> — same value: 1 nF, 50 V, X7R, 0805                                                                                                                                                                            |

## 2. The four out-of-stock items — what to do

| Item | Ref | Original                   | Our decision                                                                                                                                                                                                                                                                                                                                                 |
|------|-----|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | U2  | 4091 / D36V50F5 (Pololu)   | <b>Please source and fit it</b> — see §3 below. Do not substitute another regulator.                                                                                                                                                                                                                                                                         |
| 13   | U1  | TEENSY41 (PJRC Teensy 4.1) | <b>Customer supplied.</b> We buy the Teensy 4.1 ( <b>with pins</b> ), so nothing needs soldering on the module. <b>Please fit the two 1×24 2.54 mm female sockets</b> into the U1 footprint (Samtec SSW-124-01-G-S, or any standard 2.54 mm 1×24 socket — say the word and we will consign the Samtec parts). We plug the Teensy into the sockets ourselves. |
| 11   | D2  | SS14 (Diodes Inc)          | <b>Please use onsemi SS14</b> (LCSC C83852, stock >100 000). Alternative: Vishay SS14-                                                                                                                                                                                                                                                                       |

| Item | Ref | Original | Our decision |
|------|-----|----------|--------------|
|------|-----|----------|--------------|

E3/61T (LCSC C47460). Identical 40 V / 1 A SMA Schottky.

|    |    |                 |                                                                                                                                                                                                                                                                                                                   |
|----|----|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 28 | D5 | PESD3V3U1UL,315 | <b>Please use Nexperia PESD5V0F1BL,315</b> (LCSC C45961, stock >50 000) — same SOD-882 / DFN1006-2 land pattern. Alternative if unavailable: PESD5V0U1UL,315 (LCSC C85401).                                                                                                                                       |
|    | F3 |                 |                                                                                                                                                                                                                                                                                                                   |
|    | F4 | 1206L010/30YR   |                                                                                                                                                                                                                                                                                                                   |
| 48 | F5 |                 | <b>This part number does not exist</b> in the Littelfuse 1206L series (the 0.10 A device is only made as 1206L010/60, i.e. 1206L010/60WR). <b>Please use 1206L010/60WR</b> (LCSC C2153714, stock ~7 000; also Newark 10 764 / Digi-Key 4 861). Same 1206 package, same 100 mA hold current, 60 V instead of 30 V. |

### 3. U2 — the Pololu D36V50F5: please purchase and fit it

You told us this part is out of stock and asked us to recommend a substitute or supply it. **Please source it instead — we cannot substitute it** (it is the main 5 V rail and the PCB footprint is specific to this module).

It **is** available today:

| Supplier         | Part number                                   | Stock            | Price                          |
|------------------|-----------------------------------------------|------------------|--------------------------------|
| Digi-Key         | <b>2183-4091-ND</b> (Pololu 4091)             | 264 pcs          | US\$39.95                      |
| Pololu (factory) | <b>item 4091</b> ,<br>pololu.com/product/4091 | made to<br>order | US\$39.95 (US\$36.75 at qty 5) |

You already sourced our **Amphenol 901-143** SMA jack and our **Keystone 1058** battery holder for this same quotation, and neither of those is an LCSC part either — please source this module the same way. **\*\*We will pay the part price plus your normal sourcing fee.\*\*** If for any reason you cannot purchase it from an outside supplier, tell us before the quote is finalised and **\*\*we will ship five of them to you as consigned parts\*\*** — but please try to source it first, as consignment from the US costs both of us time and customs paperwork.

#### 4. The other two notes

| Item | Note                                                | Our reply                                                                                                                                                                                                            |
|------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 14   | BT1 1058 — "quoted part number is 1058, not CR2032" | <b>1058 is correct.</b> It is the Keystone SMD 20 mm coin-cell holder. "CR2032" in the description is the <b>cell</b> , which is not part of the BOM. Please supply the holder; <b>do not fit the cell</b> (see §6). |
| 24   | U8 NEO-M9N-00B — "price is changing higher"         | Understood. <b>Please confirm the final unit price with us before production starts.</b> If it exceeds <b>US\$20 per unit</b> please stop and tell us before ordering.                                               |

#### 5. One part-name warning

Item 2, TRACO TSR 1-2433: this must be **genuine TRACO Power**. A module with the same generic name is sold on LCSC (C53183919) but the brand is \*YLPTEC\* — that is **not acceptable** as the TRACO part. Please quote and supply genuine TRACO Power from an authorised distributor (Digi-Key, TME, Newark or Sager all have it in stock).

#### 6. What we fit ourselves after assembly — please do NOT fit these

Everything below is a plug-in or hand operation, so please leave these four things to us:

- **BT1 — do not fit the CR2032 cell.** It is a primary lithium cell: no reflow, no soldering, fit after cleaning. We insert it by hand.

- **JP1 — leave the shunt off.** We fit it only after our unloaded power-rail tests have passed.

- **U1 — do not fit the Teensy.** Please fit only the two 1×24 sockets (§2). We plug the Teensy in.

- **J9 RTC lead** — we attach the short 2-wire lead from J9 to the Teensy's auxiliary VBAT/GND pads. Just fit J9 to the board.

#### 7. What is unchanged

- The **BOM structure, references, quantities and footprints** are unchanged —

every substitution above is the same package/footprint, so **\*\*no PCB change is required\*\***.

- The **bare PCB** settings from your quote are accepted as-is: 4 layers, 150.05 × 155.05 mm, 1.6 mm, 1 oz outer copper, **ENIG**, impedance control ON (50 Ω single-ended on the GNSS feed), white silkscreen, green mask, do not panelise, keep the antenna notch in the profile.

### 8. What we need back from you

- The **revised quotation** for 5 units from the attached corrected BOM, with **U2 (Pololu 4091) priced as a sourced part** and the **\*\*two 1×24 sockets for U1 added\*\*** as assembly items.
- Confirmation that the five corrected MPNs in §1–§2 are **matched and in stock** in your system, with lead times.
- The final confirmed price for **NE0-M9N-00B** before production (see §4).
- If you cannot source U2 — tell us **now**, so we can consign it in time.

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#### Summary of changes (for your BOM edit)

| Ref      | Out                         | In                                      | Who fits it               |
|----------|-----------------------------|-----------------------------------------|---------------------------|
| C10      | GRM21BR71H103KA01L (Murata) | CL21B103KBANNNC (Samsung)               | PCBWay                    |
| C11      | GRM21BR71H225KA01L (Murata) | CC0805KKX7R9BB225 (Yageo)               | PCBWay                    |
| C18      | GRM21BR71H102KA01L (Murata) | CC0805KRX7R9BB102 (Yageo)               | PCBWay                    |
| D2       | SS14 (Diodes Incorporated)  | SS14 (onsemi)                           | PCBWay                    |
| D5       | PESD3V3U1UL, 315 (Nexperia) | PESD5V0F1BL, 315 (Nexperia)             | PCBWay                    |
| F3 F4 F5 | 1206L010/30YR (Littelfuse)  | 1206L010/60WR (Littelfuse)              | PCBWay                    |
| U2       | Pololu 4091 (out of stock)  | same part, sourced from Digi-Key/Pololu | <b>PCBWay</b>             |
| U1       | Teensy 4.1 (out of stock)   | same part, supplied by us (with pins)   | sockets: PCBWay · Teensy: |

| Ref | Out | In | Who fits it |
|-----|-----|----|-------------|
|-----|-----|----|-------------|

us